

Hyesung Jeon

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Research Interests

Keywords: Efficient AI Serving, Agentic AI Systems, KV Cache Compression, PEFT, Model compression

My primary research focus is building efficient algorithms and serving systems for generative models, such as large language models, large multi-modal models, and diffusion models. My work spans the post-training stack and efficient inference, from quantization-aware fine-tuning to KV cache management for agentic LLMs. I am particularly interested in the emerging efficiency challenges with long-context inference and agentic workloads.

In particular, my research interests lie in:

- Model compression (quantization and pruning) and parameter-efficient fine-tuning
- KV cache compression and efficient attention mechanisms
- Efficient inference systems for LLM-based agents
- Hardware-software co-design for low-precision training and inference

Education

Seoul National University

Ph.D. Student in Electrical and Computer Engineering

Mar. 2023 – (Present)

Advisor: Prof. Jae-Joon Kim

B.S. in Electrical and Computer Engineering

Mar. 2019 – Feb. 2023

Graduated Summa Cum Laude (GPA 4.04/4.30)

Gyeonggi Science High School

Math and Science Specialized High School

Mar. 2016 – Feb. 2019

Work Experiences

MangoBoost

Apr. 2022 – Jun. 2022

Internship

RDMA System Architecture Design for Data Processing Unit

Mentor: Prof. Jangwoo Kim

SK Hynix Solution Center

Internship

Jun 2021 – Aug. 2021

Deep Learning Network Design for SoC-NAND Validation Indicator

The Encouragement Prize in the Internship Workshop

Mentor: Dr. Yong Lee

Seoul National University

Graduate School of Convergence Science and Technology

Jan. 2021 – Feb. 2021

Student Researcher

Deep Learning Network Design for Nano-optical Layer Architecture

Mentor: Prof. Changsoon Kim

Student Researcher

End-to-End Development of an English Subtitle Generation Web Service

End-to-End Development of a Visitor Counter Application with a Face Detection Model

Mentor: Dr. Seokkyu Kwon

Publications

- [9] **Hyesung Jeon**, Hyeongju Ha, Seoyoung Lee, Beomseok Kang, Jae-Joon Kim, "KVCMAS: Efficient KV cache Correction for Shared Context in Multi-Agent Systems", arXiv:2609.34060, Sep. 2026.
- [8] **Hyesung Jeon**, Hyeongju Ha, Jae-Joon Kim, "PReCache: Efficient KV Cache Sharing for Multi-LoRA Agents via Low-Rank Precomputation and Neutral Reconstruction", arXiv: 2609.34054, Sep. 2026.
- [7] **Hyesung Jeon**, Hyeongju Ha, Jae-Joon Kim, "LRAgent: Efficient KV Cache Sharing for Multi-LoRA LLM Agents", The 43rd International Conference on Machine Learning (ICML), Jul. 2026.
- [6] **Hyesung Jeon***, Seojune Lee*, Beomseok Kang, Yulhwa Kim, Jae-Joon Kim, "QWHA: Quantization-Aware Walsh-Hadamard Adaptation for Parameter-Efficient Fine-Tuning on Large Language Models", The 14th International Conference on Learning Representations (ICLR), April. 2026.
- [5] Jehun Lee, **Hyesung Jeon**, Juchan Lee, Jae-Joon Kim, "PRESTE: Preserving Tiny Exponent Precision for Efficient Sub-8-bit LLM Inference and Fine-Tuning", The 45th International Conference on Computer Aided Design (ICCAD), Nov. 2026.
- [4] **Hyesung Jeon**, Yulhwa Kim, Jae-Joon Kim, "L4Q: Parameter Efficient Quantization-Aware Fine-Tuning on Large Language Models", The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Jul. 2025.
- [3] Yulhwa Kim, Dongwon Jo, **Hyesung Jeon**, Taesu Kim, Daehyun Ahn, Hyungjun Kim, Jae-Joon Kim, "Leveraging Early-Stage Robustness in Diffusion Models for Efficient and High-Quality Image Synthesis", The 37th Conference on Neural Information Processing Systems (NeurIPS), Dec. 2023.
- [2] Jiwoong Choi, Minkyu Kim, Daehyun Ahn, Taesu Kim, Yulhwa Kim, Dongwon Jo, **Hyesung Jeon**, Jae-Joon kim, Hyungjun Kim, "Squeezing Large-Scaling Diffusion Models for Mobile", The 40th International Conference on Machine Learning (ICML) Workshop on Challenges of Deploying Generative AI, Jul. 2023.
- [1] **Hyesung Jeon**, Jae-Joon Kim, "AND-Net Based Multi Precision Neural Network Accelerator Design", B.S. Graduate Paper, Feb. 2023.

Honors and Awards

Qualcomm Innovation Fellowship Korea (Winner) L4Q: Parameter Efficient Quantization-Aware Fine-Tuning on Large Language Models	2025
Samsung SAIT Computer Engineering Challenge (3rd Prize) vLLM-based LLM Inference Acceleration on Multi-GPU Systems	2023
SNU Social Responsibility+ Competition (The Encouragement Prize)	2020
SNU Electrical Circuit Design Workshop (The Excellence Prize)	2020
Korea Presidential Science Scholarship (Full Tuition Scholarship)	2019 – 2023
Intel International Science and Engineering Fair (Finalist)	2018

Invited Talks

Cortiq Summit: Agentic AI (Speaker)	2026
Efficient AI Meetup Korea (Speaker)	2024, 2026

Skills

Language	English, Korean
Programming Language	Python, C/C++
Deep Learning Frameworks	PyTorch, vLLM, SGLang, CUDA, Triton, Verilog, HLS

Leadership & Volunteering

Reviewer & Committee

ICLR (2026), ICML (2026), NeurIPS (2026), ARR (2026), TCAD (2026) Reviewer

SNU Tomorrow's Engineers Membership (Chairman)

2021-2023

Organized and participated undergraduate mentorship and scholarship programs.

References

Prof. Jae-Joon Kim

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Seoul National University
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